Week 5. By Deng Yufan.

Prob 1. Prove $\forall c_1 > c_0 \geq 1$. $TIME(n^{c_0}) \subsetneq TIME(n^{c_1})$.

Since $n^{c_1} > n \log n$ when $c_1 > 1$, and n^{c_1} can be easily calc efficiently, we have

$TIME(n^{c_1} / \log n^{c_1}) \subsetneq$ ~~TIME~~ $TIME(n^{c_1})$.

We have $O(n^{c_1} / \log n^{c_1}) = O(n^{c_1} / c_1 \log n) = O(n^{c_1} / \log n)$,

so $O(n^{c_0}) \subseteq O(n^{c_1} / \log n)$.

Thus $TIME(n^{c_0}) \subseteq TIME(n^{c_1} / \log n) \subsetneq TIME(n^{c_1})$.

Prob 2. a. Show P is closed under union. ~~[illegible]~~ concatenation. complement.

For algo. $M_1, M_2 \in P$, we construct M_3: for any input w:

Step 1. Sim. M_1 with w. If M_1 accept. then accept.

Step 2. Sim M_2 with w. If M_2 accept. then accept.

Step 3. If both reject. then reject.

Then $L(M_3) = L(M_1) \cup L(M_2)$, $M_3 \in P$. Then cons. M_4 for any input w:

Step 1: For all ~~possible~~ possible division for w into 2 parts xy:

Step a. Sim M_1 with x. If M_1 reject. continue to the next division.

Step b. Sim M_2 with y. ~ reject. continue.

Step c. If both accept. then accept.

Step 2. If never accept, then reject.

Then $L(M_4) = L(M_1) L(M_2)$, $M_4 \in P$ since Step 1. only loop $n+1$ times.

Cons. M_5 for any input w:

Step 1. Sim. M_1 with w. If M_1 accept. reject. If M_1 reject. accept.

Thus $L(M_5) = \overline{L(M_1)}$. $M_5 \in P$.

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b. Show P is closed under star.

For algo. $M \in P$. Cons. M', for any input w:

Step 1. Create a boolean array $b[0, 1, 2, \cdots, n]$. where $n = |w|$.

$b[0]=1$, $b[i]=0$ for $i \neq 0$ at first.

Step 2. Let virable i go through from 0 to $n-1$:

Step a. If $b[i]=0$, then continue to $i+1$.

Step b. Let virable j go through from $i+1$ to n:

Step α. Sim M with $w[i+1, i+2, \cdots, j]$.

Step β. If M accept, then set $b[j]$ to 1.

Otherwise do nothing.

Step 3. If $b[n]=1$, accept. If $b[n]=0$, reject.

Easy to prove for all i that $b[i]=1$. $w[1, 2, \cdots i] \in L(M)^*$.

Since Step 2 and step b only loop atmost n times. $M' \in P$.

Prob 3. Karatsuba Algo. for $n=2^k$ bit natural num. mul. :

For mul(x, y), we write $x = a2^{k/2} + b$. $y = c2^{k/2} + d$.

then $x \cdot y = (ac)2^k + [(a+b)(c+d) - ac - bd]2^{k/2} + bd$.

Prove time complexity is $O(n^{\log_2 3}) \simeq O(n^{1.585})$.

Let $T(n)$ be mul. two n-bit natural num complexity under this algo.

Since $\times 2^k$ can use lsh. to be $O(n)$. plus and minus is $O(n)$, we have

$$T(n) = 3T(n/2) + O(n).$$

By master thm. we have $T(n) = O(n^{\log_2 3})$.

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